# UA Adaptation and Hackathon Nepal 2026

Technical Track | 16 May 2026 | Kathmandu, Nepal

Duration: Full Day (8:45 AM – 6:00 PM)

## Unified Problem Statement — All 8 Technical Groups

### UAReady Email and Domain Validation System

Build a tool or library that validates email addresses and domain names across scripts (Devanagari, Latin, Arabic, etc.) conforming to SMTPUTF8, EAI, and IDNA2008 standards delivered as a standalone module or API.

## Problem Scope

All 8 technical groups will work on the same core problem. Each group may choose their own language, framework, and scope — but the deliverable must address the UAreadiness of email address and domain name handling. Groups are encouraged to differentiate through the script or language they target, the platform they integrate with (web form, API, CLI, backend, mobile), and the depth of standards compliance achieved.

## Core Requirements

- Accept and validate internationalised email addresses (e.g. राम@नेपाल.नेपाल) as valid input
- Validate domain names including IDNs using IDNA2008 and UTS#46 standards
- Handle Unicode normalisation (NFC) for consistent string comparison
- Conform to SMTPUTF8 (RFC 6531) and EAI (RFC 6532) for email address handling
- Reject invalid inputs gracefully with clear, localised error messages

## Stretch Goals (Optional)

- Support for multiple scripts beyond Devanagari (Arabic, Chinese, Cyrillic, Tamil)
- Integration with a live email sending test (SMTP with UTF8 support)
- Browserbased demo or embeddable widget for realtime validation
- DNS lookup to verify IDN domain resolution and MX record compliance

## Rules and Guidelines

- All code must be written on the day — opensource libraries and UA toolkits are permitted
- Each group submits as a team. No individual submissions
- Any programming language or stack may be used
- Recommended libraries: libicu, SMTPUTF8 patches, EAI validators, IDNA Python and JS packages
- Plagiarism or prebuilt solutions result in immediate disqualification
- Mentors may guide but may not write, debug, or contribute code

## Evaluation Criteria

- **UA Standards Compliance (30%).** Correct SMTPUTF8, EAI, and IDNA2008 implementation demonstrated with real test cases.
- **Technical Execution (25%).** Code quality, architecture, completeness, and a working demo.
- **Script Coverage (20%).** Range of scripts supported beyond basic Latin, with Devanagari as priority.
- **Nepal Relevance (15%).** Applicability to Nepal's digital identity landscape including .नेपाल domains and Nepali email addresses.
- **Presentation (10%).** Clarity of demo, team communication, and README documentation.

## Deliverables and Submission

- Working tool, module, or API — importable or accessible via URL or CLI command
- Public source code repository (GitHub or GitLab) — must go public by 4:30 PM
- README covering setup instructions, standards implemented, scripts supported, and known limits
- Test suite with at least 5 valid and 5 invalid internationalised email and domain inputs
- Slide deck of maximum 5 slides: problem, approach, standards compliance, demo, and impact
- 5minute team presentation followed by 3minute jury Q&A

## Resources Provided

- Highspeed WiFi, dedicated power outlets, and group workstations
- ICANN UA readiness tools, RFC documentation, and printed SMTPUTF8, EAI, and IDNA reference cards
- Test email accounts with SMTPUTF8capable mail servers for live testing
- Access to domain and email standards mentors and UA subjectmatter experts
- Lunch, tea breaks, and refreshments throughout the day

*"The same email address. The same domain. Working everywhere — in every language, every script."*